

An Agentic, Retrieval-Augmented Reading Companion for Secondary-School Reading Comprehension

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Abstract—This paper presents the second phase of a project developing an AI-based reading companion for secondary-school students with reading-comprehension difficulties. The project began as a comparative study of Retrieval-Augmented Generation (RAG) enrichment strategies over a small literary corpus; following instructor feedback, it was reformulated into a deployed, tool-calling LLM agent that reuses a redesigned retrieval pipeline as the engine behind a question-answering tool. The agent tracks a student’s reading position within a narrative work and gates its behavior accordingly, preventing it from revealing plot developments the student has not yet reached. We report the finalized system methodology and findings from a completed round of user testing with real students ($n = 6$, meeting the planned minimum of five), which surfaced both encouraging reception and concrete usability limitations.

Index Terms—retrieval-augmented generation, LLM agents, reading comprehension, anti-spoiler retrieval, educational technology

I. INTRODUCTION

Reading comprehension is a foundational literacy skill, yet a substantial share of secondary-school students engage with assigned narrative texts (“Plan Lector” readings, in the Peruvian curriculum) with limited independent comprehension support outside the classroom.

General-purpose conversational assistants paired with a document are a poor fit for this setting: they have no notion of what a specific student has already read, they risk revealing plot developments the student has not yet reached, and they offer no structured mechanism for comprehension practice or non-punitive feedback. Addressing these gaps requires an agent that is aware of a reader’s position within a text and that can adapt its behavior accordingly.

This project originated as a comparative evaluation of chunk-enrichment strategies for RAG over a small corpus of literary texts [1], finding that contextual per-chunk summaries improved retrieval recall and Mean Reciprocal Rank (MRR) over a naive baseline. Following instructor feedback that this remained an isolated experiment rather than an application, the project was reformulated (Phase 2) into a deployed system in which a redesigned retrieval pipeline is repurposed as the engine behind a question-answering tool inside a larger, router-based LLM agent.

Section II reviews related work motivating the system’s architecture. Section III describes the finalized methodology.

Section IV reports preliminary results from ongoing user testing. Section V concludes and discusses limitations.

II. STATE OF THE ART

A. Agentic LLM architectures with tool-calling

Agent frameworks that interleave reasoning traces with actions, such as ReAct [2], and that route sub-tasks to specialized tools, as in ToolLLM [3] and HuggingGPT [4], establish the router-plus-tool pattern adopted in our system: a lightweight model classifies user intent and delegates execution to a dedicated tool, while a separate, larger model performs content generation. This is consistent with the broader taxonomy of autonomous LLM agents surveyed by Wang et al. [5], and is conceptually related to RouteLLM [6], which routes between models of different strength to balance cost and quality.

B. RAG for narrative/literary question answering

Our question-answering tool builds on Retrieval-Augmented Generation [7], which grounds generation in retrieved passages to reduce hallucination. Gao et al. [8] situate our design within the progression from naive to advanced RAG. Early iterations of the pipeline experimented with Hypothetical Document Embeddings [9] (“HyDE”)—prepending LLM-generated hypothetical questions to each chunk before embedding, to close the vocabulary gap between a student’s informal query and formal literary prose—though this augmentation was later removed from the finalized pipeline (Section III-B). Retrieval uses multilingual E5 embeddings [10], selected for their competitiveness on non-English text. The FEVER task formulation [11] is methodologically relevant to our practice of returning character-offset citations alongside generated answers.

C. Automatic question generation and formative feedback

For comprehension checkpoints and answer evaluation, our design draws on two adjacent bodies of work. The roadmap for unifying LLMs with structured knowledge sources [12] supports grounding question generation in a structured text representation. In answer evaluation, automated scoring has traditionally emphasized holistic or punitive grading [13]; more recent work explores LLM tools that generate quizzes aligned with Bloom’s taxonomy [14], supporting our decision

to organize questions by cognitive type. An initial, AI-assisted literature search returned two further references for this cluster that, on full-text verification, proved unrelated to their claimed topics; we discarded both and treat non-punitive formative feedback for struggling readers as an open literature gap for this project.

D. Character-network extraction from narrative text

Character-network extraction from literary text is addressed by the modular Renard pipeline [15], and a follow-up study on how NLP sub-tasks affect network quality [16] informs our (still unimplemented) character-graph tool. Canonical named-entity recognition under-recognizes characters referred to by alias or indirect description [17]. Bamman et al. [18] present a Bayesian model of latent literary character types that is a foundational reference for structured character representation.

E. Progressive disclosure and anti-spoiler systems

Progressive disclosure of narrative content is addressed by two EMNLP 2023 papers. Wang et al. [19] detect spoilers in movie reviews using external knowledge and user networks. More directly relevant, Tran et al. [20] frame spoiler detection as depending explicitly on a viewer’s progress through a show—closely paralleling our own mechanism, in which a chunk’s spoiler status is a function of a reader’s monotonic progress index rather than an intrinsic property of the chunk, implemented at the retrieval level rather than via a learned classifier.

F. Auxiliary multimodal support

As a secondary feature outside the system’s core contribution, the agent can also generate a passage-scoped illustration using an off-the-shelf diffusion model [21], loosely motivated by dual-coding theory [22]; as this capability is peripheral to our evaluation, it is not reviewed further here.

G. Reading-progress tracking

Reading position is computed on the frontend from scroll position and expressed as a monotonic progress index over chunks. We found no directly comparable academic work on scroll-based, chunk-level position inference in reading agents, and treat this as a product/engineering decision rather than an application of prior research.

III. METHODOLOGY

A. Corpus Description

The final production corpus comprises eight narrative works assigned under the secondary-school “Plan Lector” curriculum, summarized in Table I. Three works carry a “2022” publication year in their source metadata, reflecting the digital edition used for preprocessing rather than the original publication date. In total, the corpus contains 2,353 retrievable chunks, 392 embedded comprehension checkpoints, and 12,593 underlying narrative blocks.

TABLE I
FINAL CORPUS

Work	Author	Chunks	Checkpoints
Crónica de una muerte anunciada	García Márquez	137	20
El maravilloso mago de Oz	Baum	156	9
El viejo y el mar	Hemingway	139	26
La ciudad y los perros	Vargas Llosa	675	79
La metamorfosis	Kafka	96	13
Las aventuras de Tom Sawyer	Twain	382	52
Matalaché	López Albújar	354	128
Viaje al centro de la Tierra	Verne	414	65

B. Corpus Extraction and Generation Methodology

Chunking uses a narrative-block-aware strategy (a custom NarrativeChunker) rather than fixed-size recursive splitting: consecutive narrative blocks are grouped until a token target is reached, using the tokenizer from multilingual-e5-small. Target chunk size ranges between 350 and 500 tokens, with a 100-token overlap target between consecutive chunks; the resulting corpus averages approximately 410 tokens per chunk, with zero hard truncations. Non-narrative and editorial blocks, including checkpoint markers, are excluded from chunking.

This configuration differs structurally from the fixed-size, character-based recursive chunker used in Phase 1, so Phase 1’s chunk-enrichment findings are reported here strictly as historical motivation rather than as figures that hold for the current pipeline. Unlike Phase 1, the finalized pipeline does not augment chunks with LLM-generated hypothetical questions before embedding: this HyDE-style step (Section II-B) was evaluated during development and subsequently removed in favor of embedding chunk text directly, simplifying the indexing pipeline. Chunks are embedded using multilingual-e5-base [10] into a ChromaDB vector store.

C. Training Methodology

No language or embedding model is fine-tuned in this project; all models are used off-the-shelf. Intent routing is performed by a small model (meta/llama-3.1-8b-instruct, temperature 0) that classifies each student message into one of five categories—summarize, question-answer, generate image, character graph, or unclassifiable—using the six most recent conversation turns as context. Content generation, including answering questions and generating formative feedback, is performed by nvidia/llama-3.3-nemotron-super-49b-v1.5, confirmed as the model used during the user-testing sessions reported in Section IV. A pending-checkpoint response bypasses the classifier and is routed directly to the answer-evaluation branch; an unclassifiable message triggers up to two clarification requests before falling back to question-answering.

D. Development and Deployment Methodology

The system is deployed for demonstration purposes: the frontend is a static build hosted on Vercel, and the backend

runs on a team member’s machine, exposed through a temporary Cloudflare Quick Tunnel that issues a new URL on each restart, so no permanent deployment URL exists for this project.

The router exposes five behaviors. Question-answering and checkpoint-based formative feedback are fully implemented end to end and are the primary focus of the evaluation in Section IV. On-demand illustration generation is included as a secondary, best-effort feature. Narrative summarization and a character-relationship graph remain unimplemented and return a “not yet available” notice rather than failing silently. The 392 comprehension checkpoints embedded in the corpus are static content authored offline, not generated at runtime.

E. Use of Generative AI

Generative AI is used in two capacities in this project: as the system’s core technology (the LLM-based agent under study), and as a research aid during the writing process, where an AI coding assistant helped inspect the codebase to keep this report grounded in the system’s actual state, and an initial AI-assisted literature pass informed part of Section II. References and code-derived claims produced this way were subsequently checked before being reported here; a small number of initially suggested sources did not hold up under verification and were left out rather than cited.

IV. RESULTS

User testing is complete for this report; this section covers six recorded sessions, meeting the planned minimum of five, conducted remotely via video call, with each student sharing their screen while a facilitator observed and guided the session. Five sessions used *La metamorfosis* and one used *El maravilloso mago de Oz*. Table II summarizes session-level observations.

TABLE II
USER-TESTING SESSIONS (N=6)

Session	Device	Checkpoint	Illustration	Reception
S1	Desktop	Yes	Yes (service down)	Positive
S2	Mobile→desktop mode	Yes	Yes (wrong subject)	Positive
S3	Mobile (slow)→desktop mode	Yes (after reload)	Yes (wrong subject)	Very positive
S4	Desktop	Yes	Attempted (failed)	Positive
S5	Desktop	Not stated	Yes (irrelevant image)	Positive
S6	Desktop	Yes	Not attempted	Very positive

All six participants completed at least one free-form question-answer exchange, and five of six completed a formative-feedback checkpoint (not evidenced in the available recording for S5). All six stated, unprompted, that they would use the tool for their own assigned school reading, and none reported the experience as confusing overall; one participant explicitly praised the comprehension checkpoints themselves as helpful, separately from the free-form chat.

None of the participants located the comprehension-checkpoint trigger without facilitator guidance, consistent with the system’s lack of onboarding (Section III-D). One participant answered a checkpoint question tersely and was gently encouraged by a facilitator to elaborate further; the reassurance

that there is no right or wrong answer was reiterated across sessions and appeared to reduce hesitation.

Illustration generation was the least reliable feature observed across all six sessions: it was unavailable in one session, failed to generate at all in another, and in the remaining attempts produced an image without a clear correspondence to the requested content. While debugging the feature live during one session, facilitators also noted that repeated requests with different prompts appeared to return the same cached image, suggesting a possible caching issue in addition to the content-fidelity problem observed earlier. As this remains a secondary feature of the system, we do not investigate it further here, but note that its reliability has now been an issue in essentially every attempt across two independent testing rounds.

Two concrete, user-driven feature requests emerged directly from testing rather than from the development team. First, when asked what else they might want to request from the chat, one participant (prompted by a parent) asked for a narrative summary of the book, directly matching the system’s current, unimplemented “Resumir” tool (Section III-D). Second, facilitators explicitly tested whether the chat could define an unfamiliar word the student had encountered (“*agregado*”); the agent was unable to answer, even though a different participant in an earlier round had spontaneously used the chat for the same purpose. This indicates that vocabulary support is, at best, an unreliable emergent behavior of the current QA-RAG tool rather than a dependable capability. Separately, one participant suggested letting users request that additional titles be added to the catalog.

One session also surfaced a minor logistics deviation: a participant originally assigned to *El maravilloso mago de Oz* was, at the facilitator’s discretion, redirected to *La metamorfosis* instead; this is noted here for transparency and did not appear to affect the session’s outcome.

Two of six participants joined on a mobile phone and experienced layout friction consistent with the system’s known lack of mobile optimization, both requiring a “desktop site” browser workaround; one of them also experienced an unplanned page reload mid-session that altered checkpoint state, a real-world instance of the documented session-persistence limitation. No further mobile-related issues were observed in the remaining, desktop-based sessions.

These findings should be interpreted with appropriate caution: although the target sample size was met, sessions were a facilitator-assisted convenience sample conducted over video call rather than the in-person, minimally-assisted conditions originally planned, and most sessions required substantial live technical assistance that would not be available in unsupervised, real-world use.

V. CONCLUSION

The project succeeded in reformulating a RAG chunk-enrichment comparison (Phase 1) into the retrieval engine of a deployed, tool-calling reading-comprehension agent (Phase 2), directly addressing the instructor’s feedback that Phase 1 remained an isolated experiment rather than an application.

The anti-spoiler mechanism is, to our knowledge, an original adaptation of prior spoiler-detection research [19], [20] to an educational context. User testing ($n = 6$, meeting the planned minimum) suggests a genuinely positive reception and confirms the system’s core value proposition—resolving reading doubts on demand and supporting comprehension through its checkpoint questions—while also surfacing concrete usability gaps: a lack of onboarding, unreliable illustration generation, limited mobile support, and unmet student demand for narrative summarization and vocabulary support.

Several limitations were identified over the project, some resolved during development and some still open. Resolved: the chunk-tracking defect that made a large share of chunks invisible to the frontend’s progress tracker in most books has been fixed. Open: two of five agent behaviors (summarization and the character graph) remain unimplemented stubs, despite summarization being a feature students explicitly asked for during testing; the QA-RAG tool does not reliably answer vocabulary-definition questions; illustration generation was unreliable across essentially every attempt observed, for a mix of availability, content-fidelity, and possibly caching reasons; no system-level logging of student interactions was implemented, so all evaluation evidence depends on manually recorded sessions; the character/location panel remains available for only one of the eight books; and the deployment has no permanent URL and, as an academic prototype, does not yet implement production-grade access controls. Future work includes implementing narrative summarization given the direct student demand observed, improving vocabulary-query handling, investigating the illustration tool’s caching behavior, and considering a user-facing mechanism to request additional catalog titles, as suggested by a participant.

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